Costain Nachuma

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Professional Summary

PhD student in Computer Science (4.0 GPA) researching AI-assisted software engineering: AI coding agents in code review, LLM-grounded code comprehension tools, and supply-chain security. 6 peer-reviewed publications (140+ citations) at MSR, ICSME, ACM SAC and Springer venues; 9+ years of prior industry experience.

Education

Idaho State University — Pocatello, ID
Ph.D. in Computer Science, Software Engineering & Cybersecurity | GPA: 4.0 — Aug 2023 – Spring 2028 (expected)
Dissertation: AI-assisted code comprehension and review. Advisor: Dr. Minhaz F. Zibran.

M.Sc. in Computer Science · *University of New Orleans*, New Orleans, LA — 2018 – 2020
B.Sc. in Computer Science · *University of Zambia*, Lusaka, Zambia — 2007 – 2011

Selected Publications

- **Nachuma, C.** & Zibran, M. When AI Teammates Meet Code Review: Collaboration Signals Shaping the Integration of Agent-Authored Pull Requests. *MSR 2026.*
- **Nachuma, C.** & Zibran, M. Comprendia: AI-Augmented Code Comprehension. *ICSME 2026*, Tool Demo Track.
- **Nachuma, C.**, Hossan, M., Turzo, A. & Zibran, M. Decoding Dependency Risks: A Quantitative Study of Vulnerabilities in the Maven Ecosystem. *MSR 2025.*

Also MSR 2024, ACM SAC 2024, ITNG 2025 (Springer) and ACM CompEd 2025. Full list: Google Scholar.

Professional Experience

Graduate Research & Teaching Assistant · *Idaho State University*, Pocatello, ID — Aug 2023 – Present

- Designed and shipped **Comprendia**, a 5-layer Eclipse IDE plugin with a dependency-graph UI and a gateway to Claude, OpenAI and Ollama (ICSME 2026).
- Built Python/DuckDB and Neo4j pipelines over 33,596 AI-agent pull requests and 14.4M+ Maven releases.
- Instructor of record, CS 3309 Advanced OOP (Fall 2025, 45 students); mentored 20+ students.

Software Engineer & Support Specialist · *National Assembly of Zambia*, Lusaka — 2012 – 2018, 2020 – 2023

- Led modernization of parliamentary web platforms; testing protocol raised pre-session error discovery to about 95%.
- Supported 500+ staff; automated network provisioning, cutting configuration time 90%.

Technical Skills

Languages	Python, Java, C#, JavaScript, SQL
Data & Tools	PostgreSQL, Neo4j, DuckDB, Pandas, Git, Docker, GitHub Actions, Linux
AI / LLMs	LLM integration (Claude, OpenAI, Ollama), prompt and context engineering
Research Methods	Logistic regression (clustered SEs), qualitative coding, mapping studies

Awards, Grants & Recognition

- **Top Graduate Oral Presentation** and **Top Graduate Poster**, ISU Research Symposium, 2026
- **Best Poster**, i-ETC 2024; **Best Oral Presentation**, ISU Research Symposium, 2024
- **Travel grants**, $6,134 total: NSF (CompEd 2025), ICSE 2025 SMeW, ISU Graduate School, ISU Dean's Fund

Service & Leadership

- **Program Committee**, MSR 2027 Mining Challenge; **Junior PC**, MSR 2027 Technical Track (2026–2027)
- **Student Volunteer**, ASE 2026; **Team Lead**, TechEquity AI Summit (50+ volunteers, 2,000+ attendees)